DATA MINING

Jobsheet Regression

Muhammad Annora Ega Prasetyo

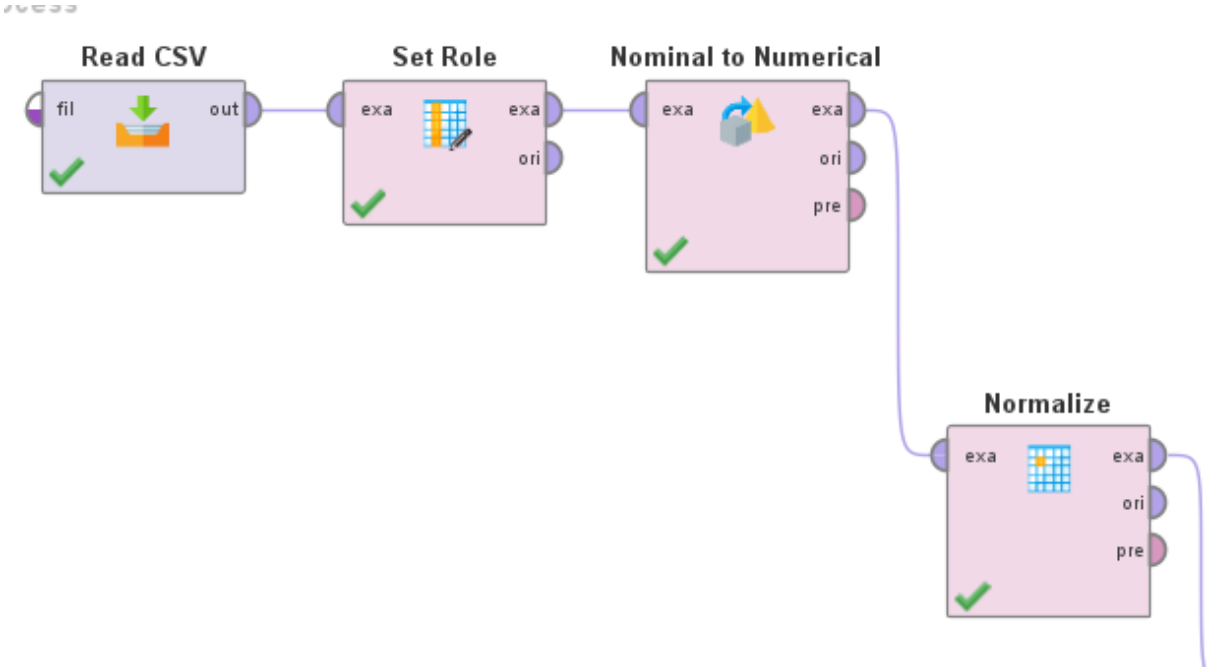
244107060069

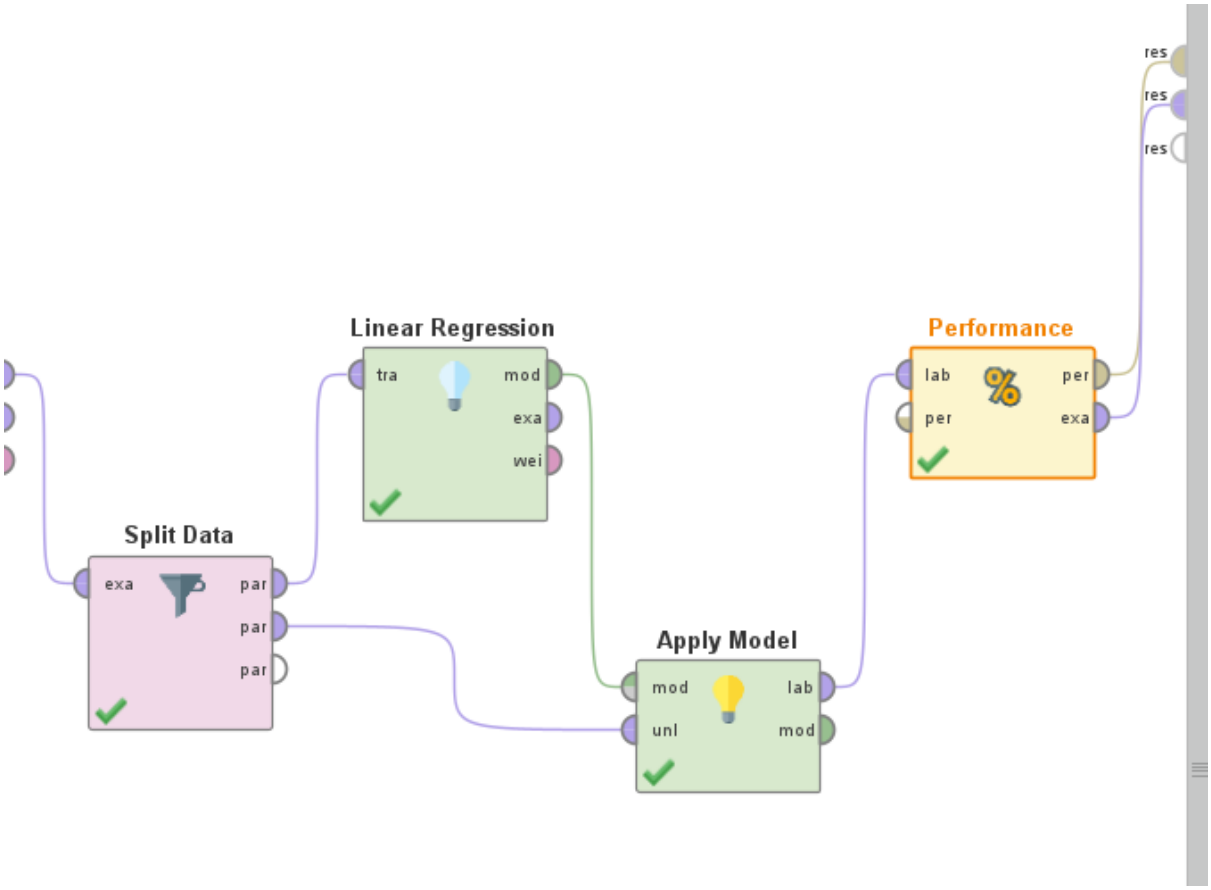
SIB 2C

1. Pilih salah satu dari dataset berikut, tuliskan deskripsi dataset, lakukan Linear Regression dan tuliskan hasil percobaan anda, termasuk fungsi regresi yang dibuat dan performa model.

(Dataset - https://www.kaggle.com/datasets/mirichoi0218/insurance)

Note: Rapid Miner





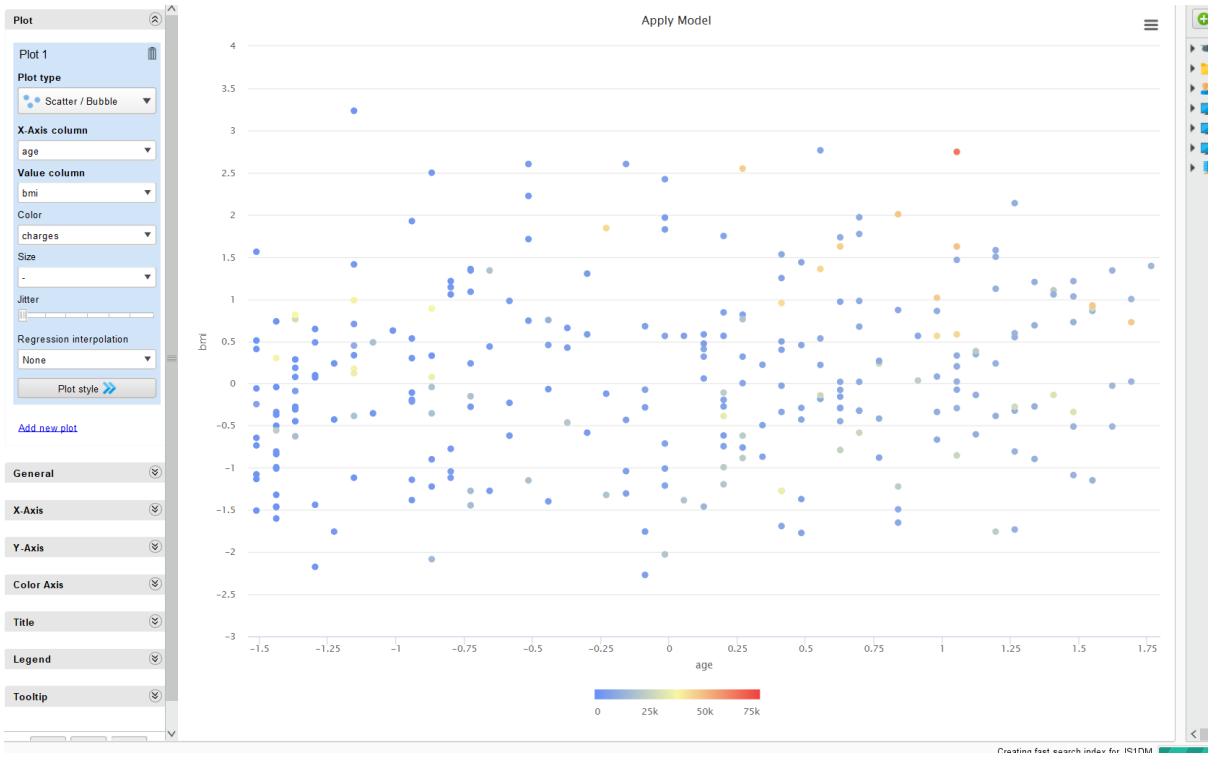
Apply Model

Row No.	charges	prediction(...	age	bmi	children	sex = female	sex = male	smoker = yes	smoker = no	region = so...	region = so...	region = no...	region = no.
1	1725.552	3545.504	-1.509	0.509	-0.079	0	1	0	1	0	1	0	0
2	11090.718	14935.545	1.195	1.502	-0.908	1	0	0	1	0	1	0	0
3	1837.237	946.235	-1.438	-0.994	-0.079	0	1	0	1	1	0	0	0
4	2395.172	2108.313	-1.154	-1.118	-0.908	0	1	0	1	0	0	0	1
5	10602.385	15439.834	1.195	1.580	-0.908	0	1	0	1	1	0	0	0
6	6203.902	7597.176	-0.157	-0.433	0.751	0	1	0	1	0	0	1	0
7	12268.632	13864.081	1.124	0.346	0.751	1	0	0	1	0	0	1	0
8	4687.797	4159.507	-1.438	-0.338	3.239	1	0	0	1	1	0	0	0
9	4949.759	8328.099	-0.584	0.978	0.751	1	0	0	1	0	1	0	0
10	3579.829	5448.138	-1.296	0.486	0.751	1	0	0	1	0	0	1	0
11	4133.642	4107.804	-0.798	-0.775	-0.079	1	0	0	1	0	0	1	0
12	11356.661	13896.341	1.266	0.549	-0.908	0	1	0	1	0	0	1	0
13	4441.213	5672.163	-0.584	-0.620	-0.079	0	1	0	1	0	0	0	1
14	7935.291	12774.118	0.412	1.250	-0.908	1	0	0	1	0	0	0	1
15	11033.662	15924.173	0.626	1.733	2.410	1	0	0	1	0	0	1	0
16	21098.554	31873.786	0.412	-1.274	0.751	0	1	1	0	0	0	1	0
17	11082.577	10936.139	1.124	-0.604	-0.908	1	0	0	1	0	0	1	0
18	30184.937	38796.206	1.409	-0.137	1.580	0	1	1	0	0	0	0	1
19	3766.884	7448.790	-0.798	1.141	-0.079	1	0	0	1	0	1	0	0
20	12105.320	12985.097	1.053	0.022	1.580	1	0	0	1	1	0	0	0
21	22412.648	33221.992	1.195	-1.757	-0.908	0	1	1	0	0	0	0	1
22	47055.532	39095.140	1.693	0.726	-0.908	0	1	1	0	0	1	0	0
23	30259.996	13244.598	1.480	-0.339	-0.908	0	1	0	1	0	0	0	1
24	1705.624	793.821	-1.509	-1.134	-0.908	0	1	0	1	0	0	0	1
25	2257.475	2660.827	-1.367	-0.277	-0.908	1	0	0	1	0	0	1	0

Performance

```
absolute_error: 4093.820 +/- 4018.534
relative_error: 44.61% +/- 40.87%
squared_correlation: 0.743
```

Hasil Visualisasi



2. Pilih salah satu dari dataset berikut, tuliskan deskripsi dataset, lakukan Logistic Regression dan tentukan performa model. (Dataset - https://www.kaggle.com/datasets/dileep070/heart-disease-prediction-using-logistic-regression)

Note: Python

CODE

a. Import library

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.linear_model import LogisticRegression
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split
from sklearn.metrics import classification_report, confusion_matrix, accuracy_score
```

b. load dataset

```
df = pd.read_csv('Dataset/framingham.csv')
```

c. Data Preprocessing

```
print("Missing Values:\n", df.isnull().sum())
df.fillna(df.median(), inplace=True)
```

d. Feature & target selection

```
X = df.drop('TenYearCHD', axis=1)
y = df['TenYearCHD']
```

e. Split data

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

f. Feature scaling

```
scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

g. Model training

```
model = LogisticRegression(class_weight='balanced', max_iter=1000)

model.fit(X_train, y_train)
```

h. Prediction

```
y_pred = model.predict(X_test)
```

i. Eval model

```
print("\n--- Model Performance ---")
print("Accuracy Score:", accuracy_score(y_test, y_pred))

print("\nConfusion Matrix:")
cm= (confusion_matrix(y_test, y_pred))
print(cm)

print("\nClassification Report:")
print(classification_report(y_test, y_pred))
```

j. Visualisasi confusion matrix

```
plt.figure()
sns.heatmap(cm, annot=True, fmt='d')
plt.title("Confusion Matrix")
plt.xlabel("Predicted Label")
plt.ylabel("Actual Label")
plt.show()
```

k. Visual distribusi target

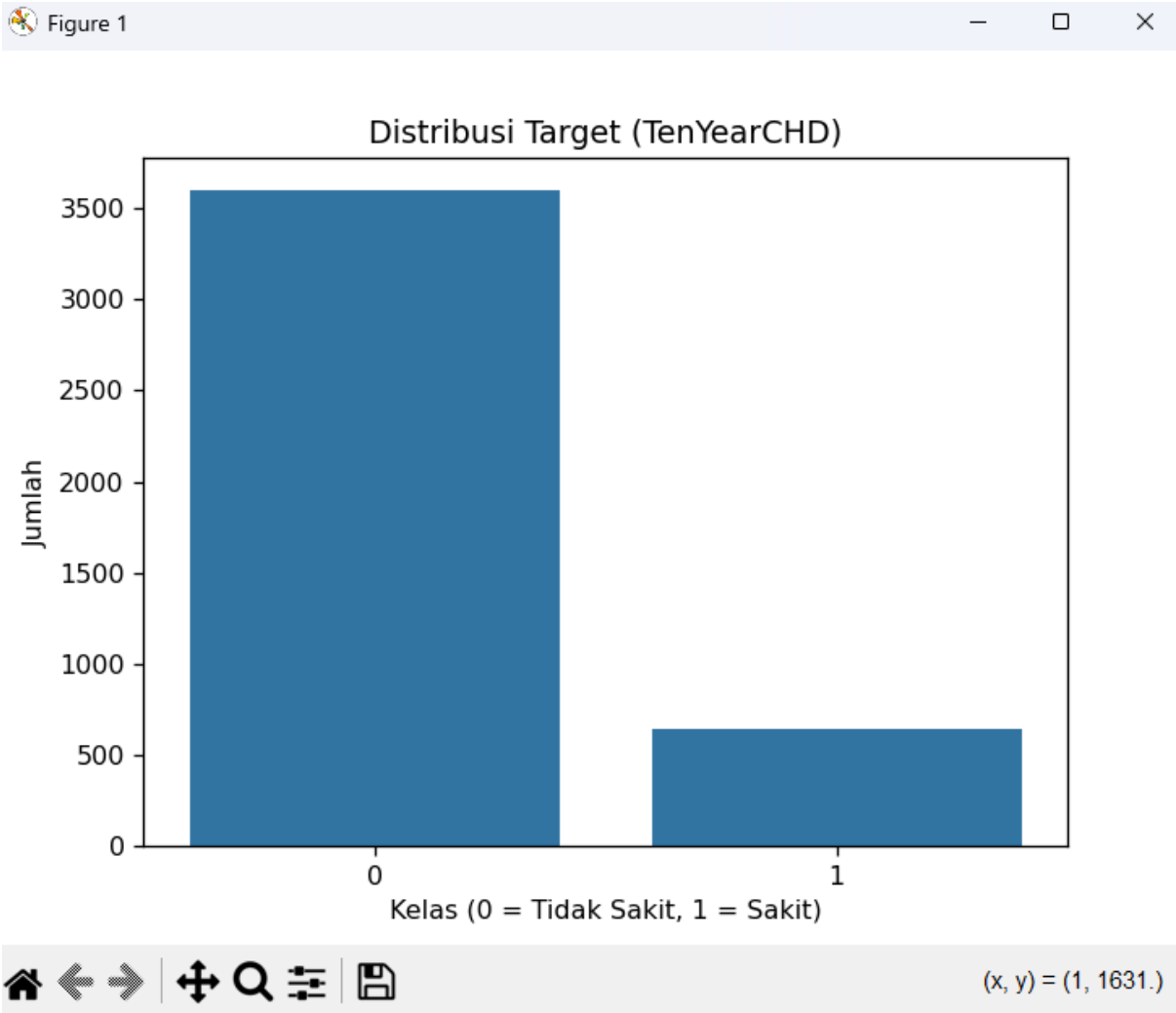
```
plt.figure()
sns.countplot(x=y)
plt.title("Distribusi Target (TenYearCHD)")
plt.xlabel("Kelas (0 = Tidak Sakit, 1 = Sakit)")
plt.ylabel("Jumlah")
plt.show()
```

OUTPUT

```
Missing Values:
 male                0
age                  0
education          105
currentSmoker        0
cigsPerDay          29
BPMeds              53
prevalentStroke      0
prevalentHyp         0
diabetes             0
totChol             50
sysBP                0
diaBP                0
BMI                 19
heartRate            1
glucose            388
TenYearCHD           0
dtype: int64
```

```
--- Model Performance ---
Accuracy Score: 0.6450471698113207

Confusion Matrix:
[[466 258]
 [ 43  81]]

Classification Report:
              precision    recall  f1-score   support

           0       0.92      0.64      0.76       724
           1       0.24      0.65      0.35       124

    accuracy                           0.65       848
   macro avg       0.58      0.65      0.55       848
weighted avg       0.82      0.65      0.70       848
```

Figure 1
Confusion Matrix
466
258
43
81
0
1
Actual Label
0
1
Predicted Label
450
400
350
300
250
200
150
100
50



3. 3. Gunakan data harga labu berikut untuk membuat prediksi menggunakan Polynomial regression dan tentukan performa model

(Dataset - https://www.kaggle.com/code/residentmario/pumpkin-price-polynomial-regression/input?select=new-york_9-24-2016_9-30-2017.csv)

Note: Python

Code

a. import library

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.preprocessing import PolynomialFeatures
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error, r2_score
```

b. load dataset

```
df = pd.read_csv('Dataset/new-york_9-24-2016_9-30-2017.csv')
```

c. Select kolom

```
df = df[['Date', 'Low Price']]
```

d. Preprocessing data

```
df['Date'] = pd.to_datetime(df['Date'])
df['Date'] = df['Date'].map(pd.Timestamp.toordinal)

df = df.dropna()
```

e. Feature & target

```
X = df[['Date']]
y = df['Low Price']
```

f. Sort data

```
df = df.sort_values('Date')
```

g. Split data

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

h. Polynomial transform

```
poly = PolynomialFeatures(degree=2)

X_train_poly = poly.fit_transform(X_train)
X_test_poly = poly.transform(X_test)
```

i. Train model

```
model = LinearRegression()
model.fit(X_train_poly, y_train)
```

j. Membuat kurva

```
X_range = np.linspace(X['Date'].min(), X['Date'].max(), 100).reshape(-1,1)

X_range_poly = poly.transform(X_range)
y_range_pred = model.predict(X_range_poly)
```

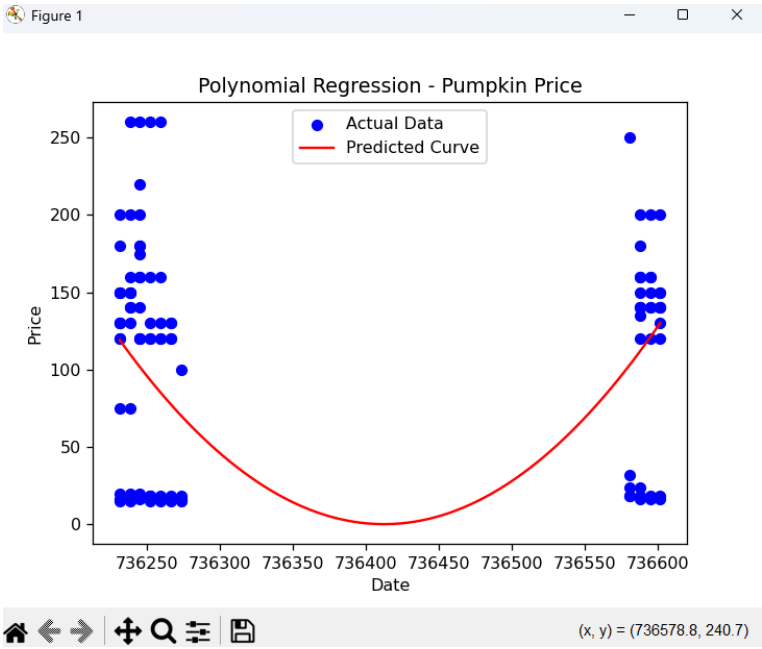
k. Prediction

```
y_pred = model.predict(X_test_poly)
```

l. Eval mode

```
mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)
```

m. visualisasi

```
plt.scatter(X, y, color='blue', label='Actual Data')
plt.plot(X_range, y_range_pred, color='red', label='Predicted Curve')

plt.title("Polynomial Regression - Pumpkin Price")
plt.xlabel("Date")
plt.ylabel("Price")
plt.legend()
plt.show()
```

Output